
Counting and conditional probability

Events · Counting · Bayes' theorem

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Probability and Statistics for Engineers

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Monday: events and counting

From the opening lesson

$$\omega \in \Omega, \quad A \subseteq \Omega.$$

For a finite model,

$$q_\omega = P(\{\omega\}), \quad P(A) = \sum_{\omega \in A} q_\omega.$$

Today

- 1 Separate overlapping events.
- 2 Count sequences and subsets.
- 3 Turn counts into probabilities.

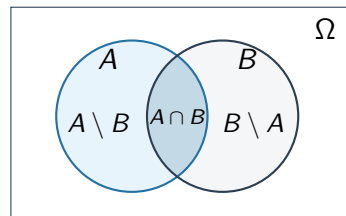
A quick check

When is $P(A) = |A|/|\Omega|$ valid?

Translate the event before calculating

Description	Event
Both A and B	$A \cap B$
A or B (or both)	$A \cup B$
Not A	$A^c = \Omega \setminus A$
B , but not A	$B \setminus A = B \cap A^c$

A union includes the overlap once.



Disjoint means $A \cap B = \emptyset$.

Probability rules and a useful complement

The rules we use

- ① $P(A) \geq 0$ for every event A .
- ② $P(\Omega) = 1$.
- ③ If $E_1, \dots, E_m$ are pairwise disjoint, then

$$P\left(\bigcup_{i=1}^m E_i\right) = \sum_{i=1}^m P(E_i).$$

The general additivity axiom also allows a countable sequence of disjoint events.

From the opening lesson

$\Omega = A \cup A^c$ is a disjoint union, so

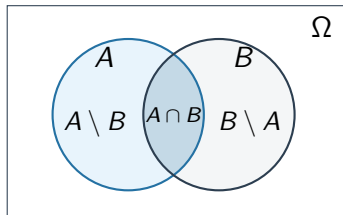
$$P(A^c) = 1 - P(A).$$

For “at least one,” first consider “none.”

Derive the two-event union rule

For any events A and B

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

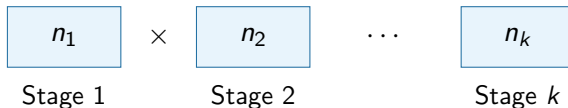


Split B into $A \cap B$ and $B \setminus A$.
Then split $A \cup B$ into A and $B \setminus A$.

Count successive choices

If there are n_1 choices at the first stage and n_2 choices after *every* first choice,

number of two-stage outcomes = $n_1 n_2$.



k ordered choices from n objects, with replacement:

$$n \cdot n \cdots n = n^k.$$

Counting outcomes does not, by itself, make them equally likely.

Check: how many length-three strings use six symbols if repetition is allowed?

Ordered selections without replacement

Choose k distinct objects from n and record their order.

Falling factorial

For integers $0 \leq k \leq n$,

$$(n)_k = n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

The empty selection has count $(n)_0 = 1$; by convention $0! = 1$.

Three different roles

From ten people, assign a president, a secretary, and a treasurer. No person holds two roles.

One committee has six ordered representations

For three distinct people a, b, c , the same committee $\{a, b, c\}$ appears as

$$\begin{array}{ccc} (a, b, c) & (a, c, b) & (b, a, c) \\ (b, c, a) & (c, a, b) & (c, b, a) \end{array}$$

There are $3 \cdot 2 \cdot 1 = 3!$ orders for each three-person committee.

The same reasoning for k distinct objects

Each k -element subset has exactly $k!$ ordered representations.

The divisor must be the same for every subset.

Derive the number of k -element subsets

Combinations

For integers $0 \leq k \leq n$,

$$\binom{n}{k} = \frac{(n)_k}{k!} = \frac{n!}{k!(n-k)!}.$$

Count the ordered selections in two ways:

Return to the ten people: how many committees of three?

Four items, six locations

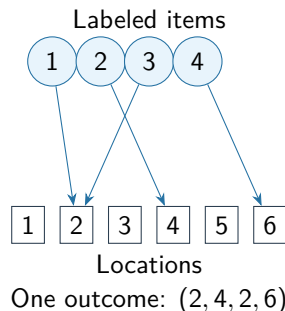
Place four labeled items into six labeled locations.

Model: each item chooses a location uniformly and independently of the others.

An outcome is an ordered tuple:

$$\omega = (x_1, x_2, x_3, x_4) \in \{1, \dots, 6\}^4.$$

Each tuple has probability 6^{-4} .



What is the probability that at least two items share a location?

An idealized allocation model, motivated by collisions in hashing.

Count no collisions, then take the complement

Let $C = \{\text{at least two items share a location}\}$.

Then $C^c = \{\text{all four locations are distinct}\}$.

Three steps

Count all ordered tuples. Count the tuples in C^c . Use equal probabilities.

The tuple is ordered by item label, even when only collisions interest us.

Practice: at least one marked item

Uniformly choose a three-element subset

There are ten distinct items, four marked and six unmarked. What is the probability that the chosen subset contains at least one marked item?

Set up your solution

- ① What are the equally likely outcomes?
 - ② Which event is easier to count: at least one marked item, or none?
-
-
-

Counting: order, replacement, and probability

Outcome	Repetition	Count
Ordered length- k sequence	Allowed	n^k
Ordered length- k sequence	Not allowed	$(n)_k$
Unordered k -element subset	Not allowed	$\binom{n}{k}$

$n \geq 1$ and $k \geq 0$ are integers; without replacement, $k \leq n$.

Before dividing counts

Specify the elementary outcomes and justify why they are equally likely.

$$P(A^c) = 1 - P(A), \quad P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Wednesday: probability after new information

Today: identify the condition, change the reference set, and calculate along a sequence of observations.

Retrieve Monday's model

Two independent fair dice are rolled. Let $A = \{\text{sum is } 7\}$.

Before any observation: list the outcomes in A and find $P(A)$.

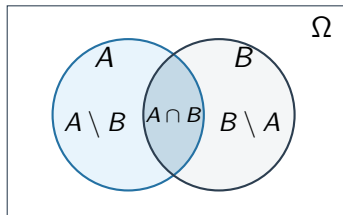
Now suppose we learn that at least one die is 4. What changes?

Conditional probability: restrict and renormalize

Definition: B is the information supplied

For $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$



Restrict: retain the outcomes in B .

Renormalize: divide their probabilities by $P(B)$ so their total becomes one.

Count A within B : only $A \cap B$ remains relevant.

Conditional weights need not be equal

For a finite model with $q_\omega = P(\{\omega\})$ and $P(B) > 0$,

$$P(A \mid B) = \frac{\sum_{\omega \in A \cap B} q_\omega}{\sum_{\omega \in B} q_\omega}.$$

Equal original weights

$$P(A \mid B) = \frac{|A \cap B|}{|B|}.$$

The common weight cancels.

First identify B . Then decide whether counting is justified.

Unequal original weights

Keep the original relative weights of the outcomes in B .

Restricting a model does not make it uniform.

Two dice: at least one die is four

		Second die					
		1	2	3	4	5	6
First die	1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
	3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
	4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
	5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
	6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

$A = \{\text{sum is 7}\},$

$B = \{\text{at least one die is 4}\}.$

The outlined row and column form B . The circled cells form $A \cap B$.

Find $|B|$, $|A \cap B|$, and $P(A | B)$.

Different information gives a different denominator

Keep $A = \{\text{sum is 7}\}$ and the same two fair dice.

Condition B	$ B $	$ A \cap B $	$P(A B)$
At least one die is 4	11	2	2/11
First die is 4	_____	_____	_____
Exactly one die is 4	_____	_____	_____

Explain: which condition excludes $(4, 4)$? Which condition excludes $(3, 4)$?

Derive the conditional product rule

For $P(B) > 0$

$$P(A \cap B) = P(B) P(A | B).$$

Start with the conditional definition and multiply by $P(B)$.

If $P(A) > 0$, we may also write

$$P(A \cap B) = P(A) P(B | A).$$

No independence assumption is needed

The second factor must use the information in the first event.

Probability rules within the condition

Fix an event B with $P(B) > 0$.

Normalization and complement

$$P(B \mid B) = 1, \quad P(A^c \mid B) = 1 - P(A \mid B).$$

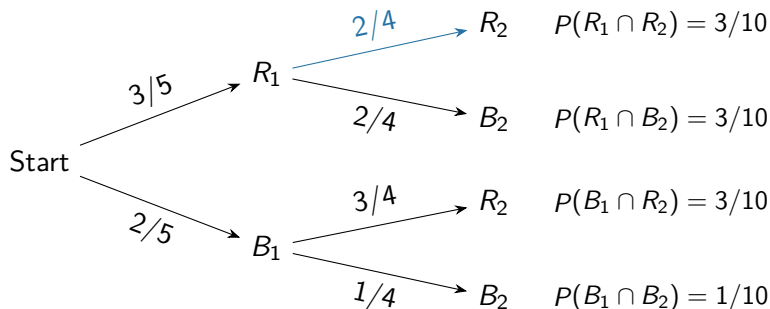
Derive both using the same denominator.

$\dot{\cup}$ denotes a disjoint union.

Two draws without replacement

Three red and two blue balls are equally likely to be drawn. Draw twice without replacement.

Let R_i and B_i mean red and blue on draw i .



Each second-stage branch is conditional on the first draw. Multiply along a path.

Explain each factor in a path probability

Find the probability of two red draws.

After a red first draw, two red and two blue balls remain.

Why not $(3/5)^2$?

The first ball is not replaced. The second factor must describe the remaining four balls.

The four complete paths are disjoint and their probabilities sum to one.

Apply the product rule twice

Three-event chain rule

If $P(A \cap B) > 0$, then

$$P(A \cap B \cap C) = P(A) P(B | A) P(C | A \cap B).$$

First treat $A \cap B$ as one event. Then expand its probability.

For red, red, blue without replacement:

$$P(R_1 \cap R_2 \cap B_3) = \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{2}{3} = \frac{1}{5}.$$

At each stage, condition on the entire preceding history.

Practice: distinguish a path from a condition

Again draw twice without replacement from three red and two blue balls.

- ① Find the probability that the two draws have different colors.
- ② Find $P(R_2 \mid B_1)$. What remains after the blue draw?

A conditional probability describes the restricted model; a path probability includes reaching that condition.

Condition, intersection, denominator

For $P(B) > 0$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(A \cap B) = P(B) P(A \mid B).$$

- 1 Name the observed event B precisely.
- 2 Identify $A \cap B$ inside that event.
- 3 Use counts only when the retained outcomes have equal weights.

Direction matters

$P(A \mid B)$ and $P(B \mid A)$ generally have different denominators.

Friday: add disjoint paths, then reverse the conditioning.

Friday: combine cases and reverse conditioning

Today

- ① Divide a model into disjoint, exhaustive cases.
- ② Compute an event probability by summing across those cases.
- ③ Update the probabilities of the cases after an observation.

Retrieve Wednesday's result

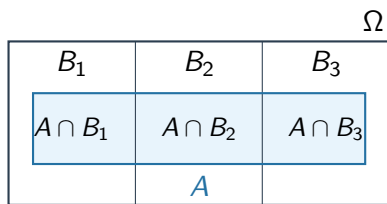
For events A and B with positive probabilities, write $P(A \cap B)$ in two ways.

A partition lists disjoint, exhaustive cases

Definition

Events $B_1, \dots, B_m$ form a partition of Ω if

$$B_i \cap B_j = \emptyset \ (i \neq j), \quad \bigcup_{i=1}^m B_i = \Omega.$$



Schematic regions; areas are not probabilities.

Every outcome belongs to exactly one case.

Check: do “first draw red” and “second draw red” partition the two-draw model?

Derive the law of total probability

Assume $B_1, \dots, B_m$ partition Ω and $P(B_i) > 0$

$$P(A) = \sum_{i=1}^m P(A \mid B_i) P(B_i).$$

Split A into disjoint pieces, then use the product rule on each piece.

The weights are $P(B_i)$, not necessarily $1/m$.

Zero-probability parts contribute zero intersection probability and may be omitted from the conditional sum.

Three sources: weight each defect probability

A randomly selected item comes from exactly one of three sources.

Let $B_i = \{\text{source } i\}$ and $D = \{\text{defective}\}$.

Source	Share $P(B_i)$	$P(D \mid B_i)$	Contribution $P(D \cap B_i)$
1	0.20	0.01	_____
2	0.30	0.02	_____
3	0.50	0.03	_____

Find $P(D)$ and explain the weights.

Adapted from DeGroot–Schervish, 4th ed., Example 2.3.4, and the earlier STAT 131 recap.

Derive Bayes' theorem

Let $B_1, \dots, B_m$ partition Ω , with $P(B_i) > 0$ and $P(A) > 0$.

Posterior probability of case B_j after observing A

$$P(B_j | A) = \frac{P(A | B_j) P(B_j)}{\sum_{i=1}^m P(A | B_i) P(B_i)}.$$

Start with the new conditioning event A .

The numerator is one case's joint contribution; the denominator adds all cases.

Among defective items, which source is more likely?

Use the contributions already computed:

$$P(D \cap B_1) = 0.002, \quad P(D \cap B_2) = 0.006, \quad P(D \cap B_3) = 0.015.$$

Given a defective item, find the probability it came from source 3.

Before observing the defect

$$P(B_3) = 0.50.$$

Both the source share and its defect probability enter the update.

After observing the defect

$P(B_3 \mid D)$ is the source-3 share among defective items.

A rare issue and an alert

A hypothetical screening model

Let $G = \{\text{genuine issue}\}$ and $L = \{\text{alert}\}$.

Genuine issues are rare	$P(G) = 0.01$
Most genuine issues trigger an alert	$P(L G) = 0.90$
Some other cases also trigger an alert	$P(L G^c) = 0.05$

If an alert occurs, what is $P(G | L)$?

$P(L | G) = 0.90$ answers a different question.

Illustrative probabilities, not measured rates from a company or dataset.

Use a common scale to see the base rate

Scale the model proportions to 10,000 cases.

	Alert L	No alert L^c	Total
Genuine issue G	90	10	100
No genuine issue G^c	495	9,405	9,900
Total	585	9,415	10,000

Restrict to the alert column and form the posterior.

Most cases have no genuine issue, so that group can supply many false alerts.
These are model-based frequencies (expected counts), not guaranteed observed counts.

Practice: infer the box from the observed color

Choose box 1 with probability $1/3$ and box 2 with probability $2/3$.
A draw is red with probability $3/4$ from box 1 and $1/4$ from box 2.

Given a red draw, which box is more likely?

Let $B_i = \{\text{box } i\}$ and $R = \{\text{red draw}\}$. Find $P(B_1 | R)$.

Compute the probability of red before reversing the condition.

Total probability and Bayes: keep the cases explicit

For a partition $B_1, \dots, B_m$ with $P(B_i) > 0$,

Combine cases

$$P(A) = \sum_{i=1}^m P(A \mid B_i) P(B_i).$$

If also $P(A) > 0$,

Update a case after an observation

$$P(B_j \mid A) = \frac{P(A \mid B_j) P(B_j)}{P(A)}.$$

Check: disjoint cases, all cases, correct weights, positive denominator.

Next week: when does conditioning leave a probability unchanged?

References for this week's material

DeGroot and Schervish

Probability and Statistics, fourth edition.

- Sections 1.5–1.8: probability properties and counting.
- Section 2.1: conditioning, multiplication, and total probability.
- Section 2.3: Bayes' theorem.

Mor Harchol-Balter

Introduction to Probability for Computing (2024).

- Section 1.5: permutations and combinations.
- Sections 2.1–2.3, 2.5–2.6: probability, conditioning, and Bayes.

Additional reference: Sheng Jiang, *STAT 131: Introduction to Probability Theory*, July 18, 2024, pp. 10–19.

Examples also adapt Antonio de Leon's earlier STAT 131 discussion sheets and recap slides.